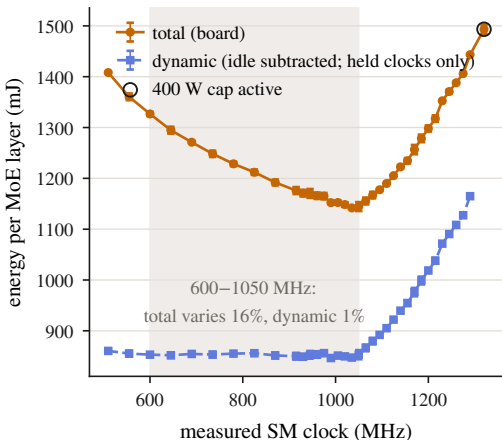


A100-SXM4-40GB, 400 W cap | Qwen3-30B-A3B MoE layer, 7744 prefill tokens, real ShareGPT routing | vLLM Triton grouped GEMM (tuned)
 6 passes per clock over a reshuffled grid, mean $\pm$ sd (the 1320 MHz point pools requests of 1335/1380/1410, $n = 18$) | total energy = NVML
 median board power $\times$ layer time; dynamic = (that power – idle measured at the same locked clock) $\times$ layer time.

Shading in (a) marks where dynamic energy is within 1% of its minimum. Per-point sd varies genuinely: 1.20 mJ at 1035 MHz against 8.75 at 1050, both $n = 6$.

(a) below the knee the rise is idle amortisation



(b) frontier, against the deployed point

